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Experiment No. 4

Aim: Creation of an account in MetaMask, and transfer of amount/cryptocurrency from one account to another.

Lab Objectives: To explore blockchain concepts by setting up a cryptocurrency wallet and performing a peer-to-peer transaction on a test network.

Lab Outcomes: Demonstrate the concept of Blockchain wallets and transactions in a real-world (test network) application.

Theory

1. What is MetaMask?

MetaMask is a non-custodial cryptocurrency wallet, available as a browser extension and mobile app, that lets a user generate and manage Ethereum-compatible accounts, hold ETH and other tokens, and interact with decentralized applications and blockchains directly from the browser. Being non-custodial means the wallet's private keys are generated and stored locally on the user's device (encrypted with a password, and recoverable via a 12-word secret recovery phrase) rather than being held by any company or exchange, so the account holder has full control of, and full responsibility for, their funds.

2. Account Creation in MetaMask

When a new MetaMask account is created, the wallet generates a public/private key pair. The private key is used to sign transactions and must never be shared, while the public key is hashed to derive the account's public wallet address (e.g., 0xbb6E6...189B4) — the identifier others use to send funds to that account. A single MetaMask installation can hold multiple accounts (all derived from the same seed phrase, or imported separately), letting one user manage several addresses from a single interface.

3. Testnets and Test Tokens

Since real ETH has monetary value, blockchain transactions during development and learning are usually performed on a testnet a separate blockchain that mimics the mainnet's behaviour but uses worthless test tokens. In this experiment, the Sepolia Testnet was used, and the account was funded with test SepoliaETH obtained from a Sepolia faucet, allowing transactions to be sent and confirmed without any real financial cost or risk.

4. How a Transaction Works

A transaction in MetaMask/Ethereum consists of a sender address, a recipient address, an amount to transfer, and a gas fee paid to the network for processing it. When a user initiates a “Send” transaction:

- The sender specifies the recipient's address and the amount to transfer.
- MetaMask estimates the network fee (gas) required to process the transaction, based on current network congestion.
- The user reviews the transaction details sender, recipient, network, amount, and fee and confirms it by signing it with their private key.
- The signed transaction is broadcast to the network, picked up by validators/miners, included in a block, and once mined, the recipient's balance is updated and the sender's balance is reduced by the amount sent plus the gas fee.

This process demonstrates the core blockchain principle of peer-to-peer value transfer without an intermediary such as a bank: the transaction is validated and recorded directly on the distributed ledger, and is publicly verifiable by anyone via a block explorer.

Procedure and Output

Step 1: Account created and funded

A MetaMask account (Account 1, address 0xbb6E6...189B4) was set up on the Sepolia Testnet and funded with 0.0600 SepoliaETH from a testnet faucet.

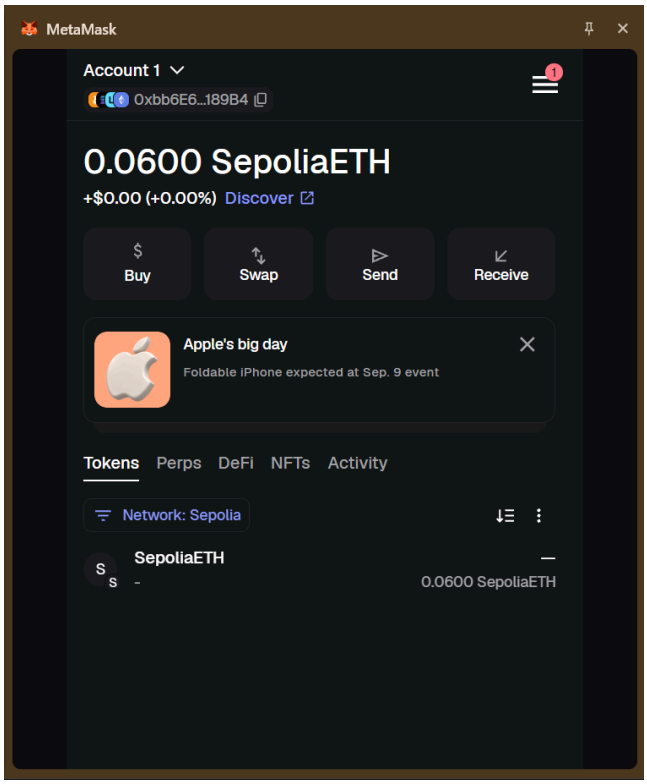


Fig 1: MetaMask wallet showing Account 1 funded with 0.0600 SepoliaETH on the Sepolia network.

Step 2: Initiating the transfer

The “Send” option was selected, the recipient address (0x78d2C...D50FA) was entered, and an amount of 0.01 SepoliaETH was specified for transfer.

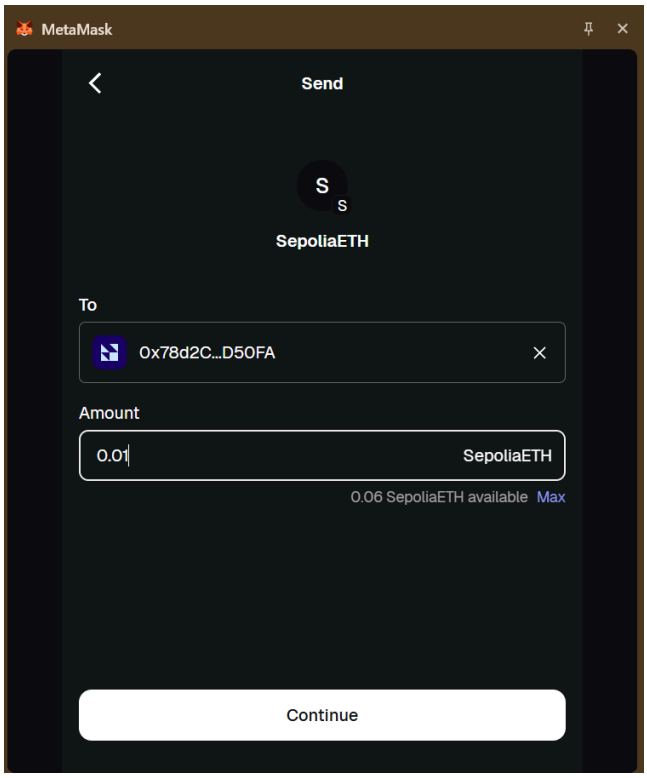


Fig 2: Send screen with recipient address and amount (0.01 SepoliaETH) entered.

Step 3: Confirming the transaction

MetaMask displayed a confirmation screen summarizing the transaction sender (Account 1), recipient address, network (Sepolia), amount (0.01 SepoliaETH), and the estimated network fee (0.0001 SepoliaETH). The transaction was confirmed and signed.

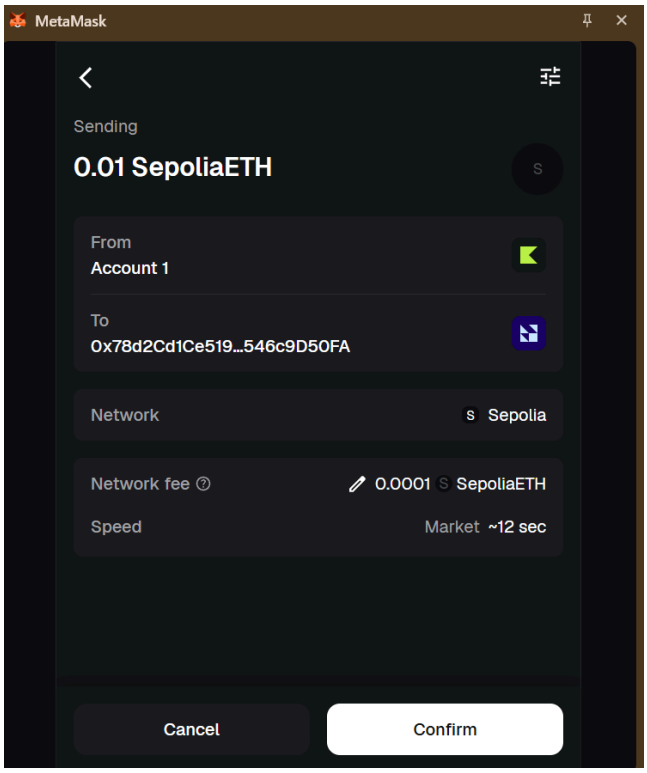


Fig 3: Transaction confirmation screen showing sender, recipient, network, and gas fee before signing.

Step 4: Verifying the updated balance

After the transaction was mined, the wallet balance updated to 0.0499 SepoliaETH reflecting the 0.01 ETH sent plus the gas fee deducted from the original 0.0600 SepoliaETH balance, confirming the transfer was successfully completed.

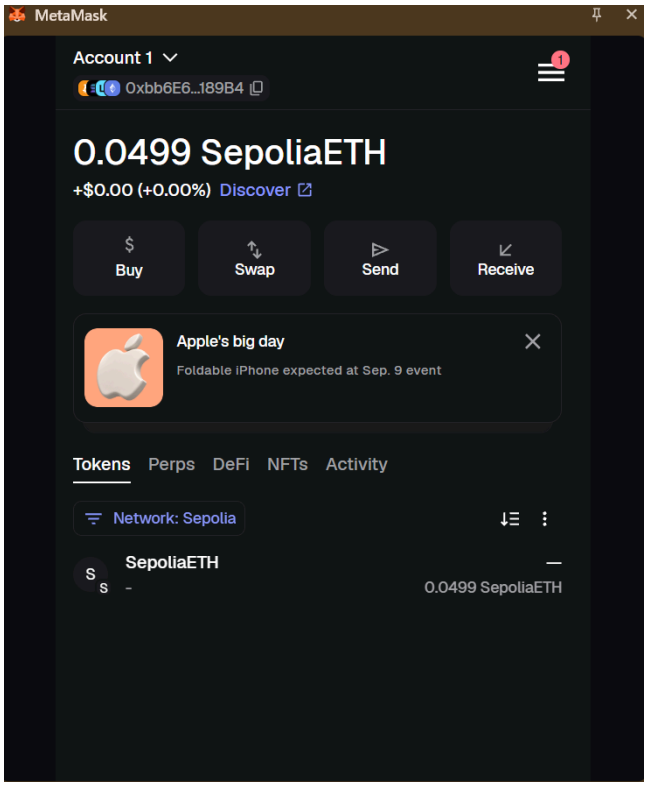
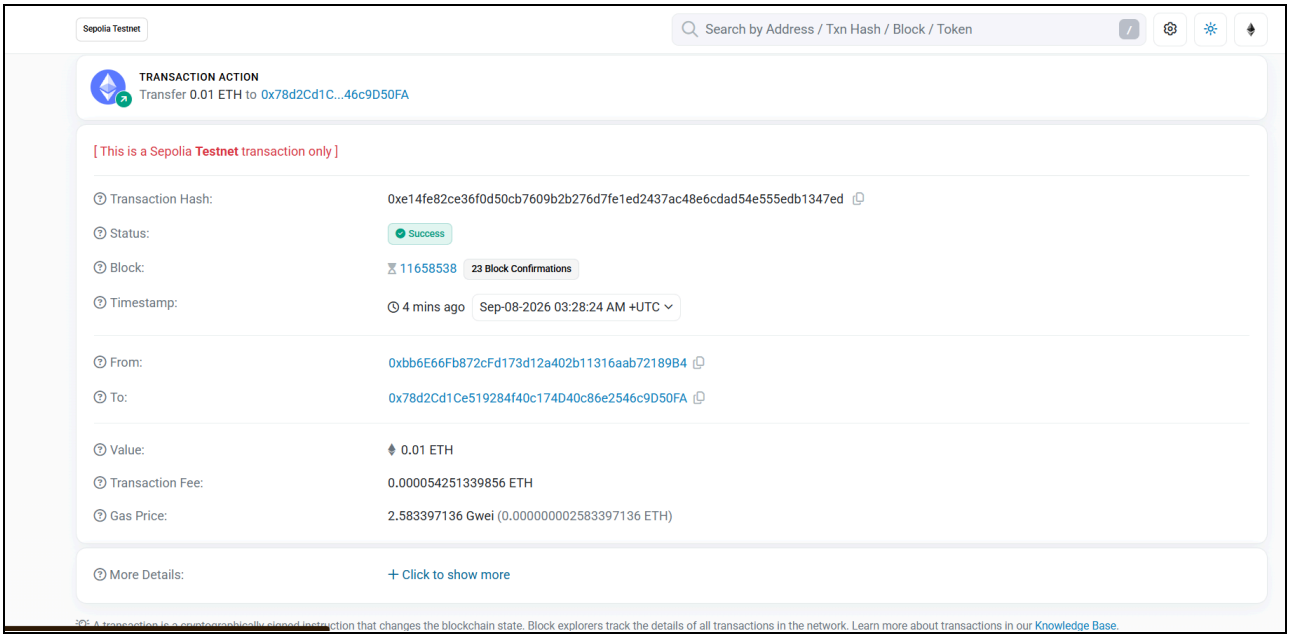


Fig 4: Updated wallet balance (0.0499 SepoliaETH) after the transaction was confirmed.

Outcome

- Understood the process of creating and managing an account in MetaMask.
- Successfully transferred SepoliaETH from one account to another on the Sepolia Testnet.
- Verified the transaction by observing the balance update before and after the transfer.



Conclusion

In this experiment, a MetaMask wallet account was created and funded with test SepoliaETH on the Sepolia Testnet. A peer-to-peer transaction of 0.01 SepoliaETH was initiated, reviewed, and confirmed through MetaMask's Send flow, and the resulting change in account balance (from 0.0600 to 0.0499 SepoliaETH, accounting for the amount sent and the network gas fee) confirmed that the transaction was successfully broadcast and mined on the network. This experiment demonstrated the fundamental blockchain workflow of wallet-based, peer-to-peer value transfer without any intermediary, and illustrated how gas fees are deducted alongside the transferred amount in every on-chain transaction.